

CSE518 Foundations of Human Computer Interaction

Lecture 1 Introduction-1

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Agenda

- What is HCI?
- Course Content
- Administrivia

Human Computer Interaction

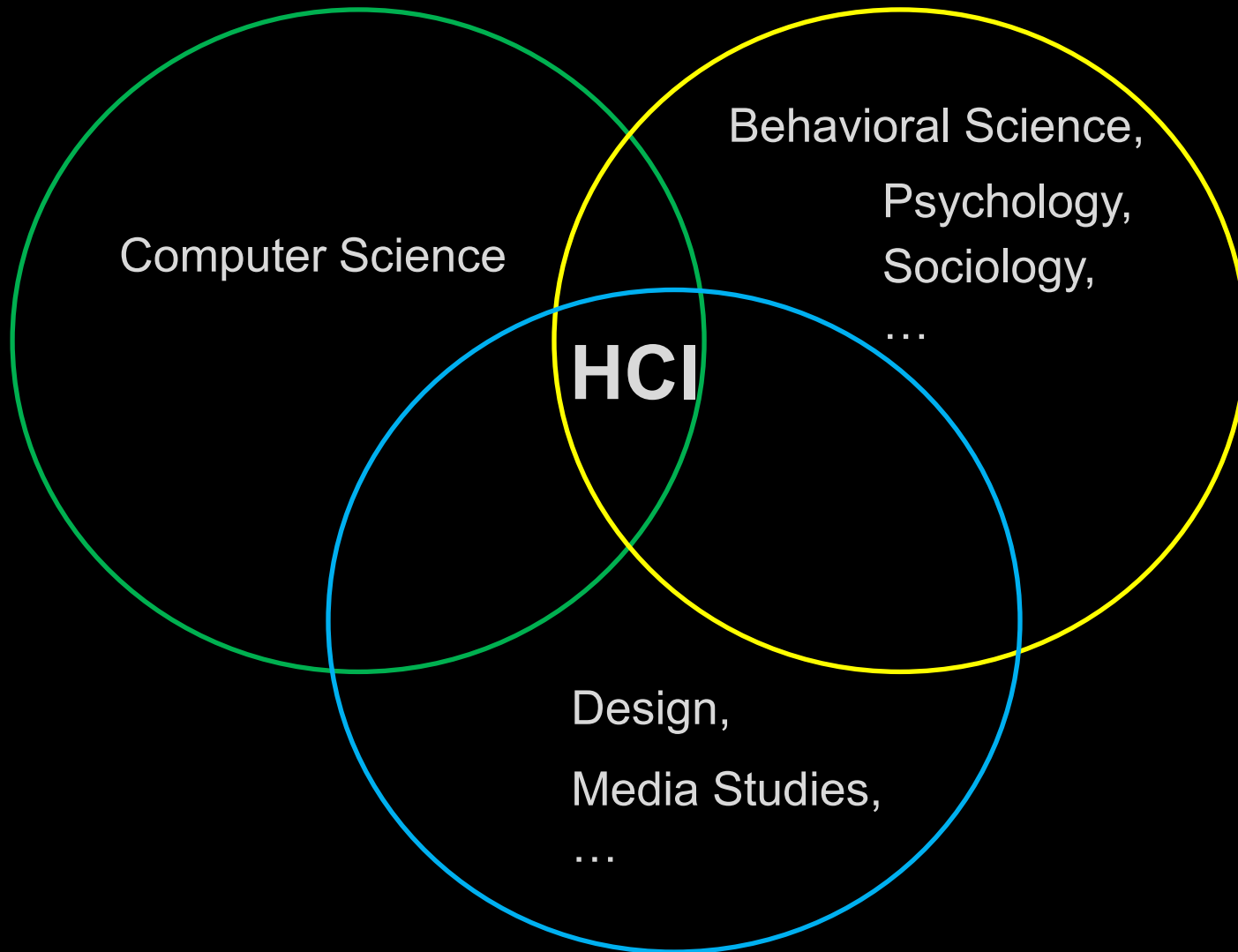
- What is Human Computer Interaction (HCI)?

A discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them.

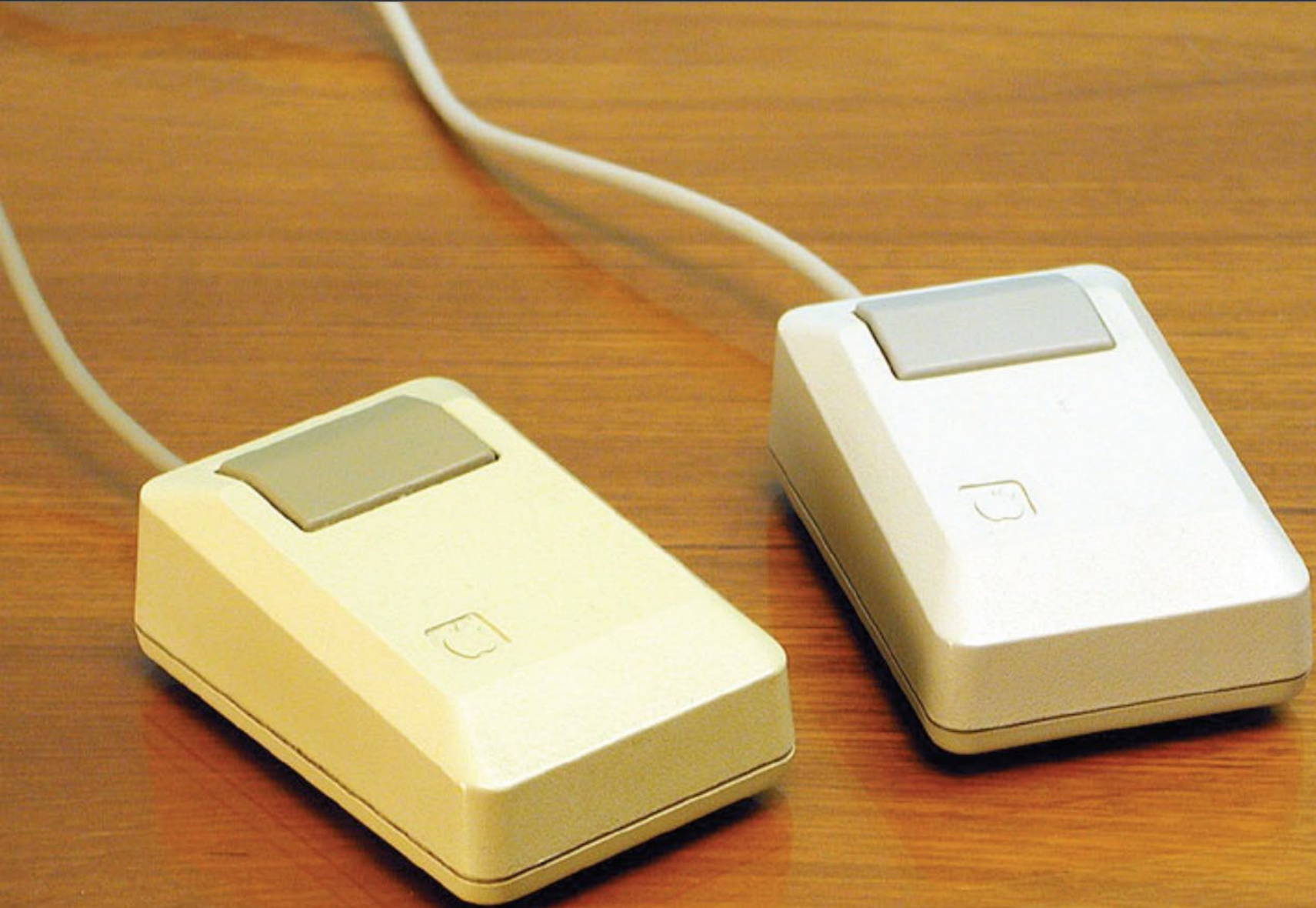
Thomas T. Hewett, Ronald Baecker, Stuart Card, Tom Carey, Jean Gasen, Marilyn Mantei, Gary Perlman, Gary Strong, and William Verplank. 1992. ACM SIGCHI Curricula for Human-Computer Interaction. Technical Report. Association for Computing Machinery, New York, NY, USA.

Human Computer Interaction

- What is Human Computer Interaction (HCI)?



Great Inventions in Human Computer Interaction





Sketch Pad Demo





GUI



Demo Videos

1982 Xerox Star



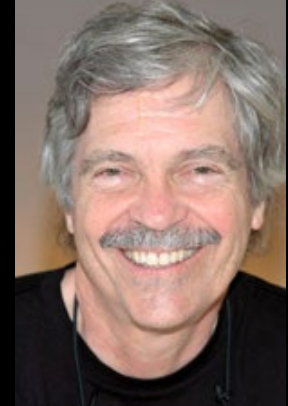
Turing Award Laureates in HCI



Ivan Sutherland



Doug Engelbart



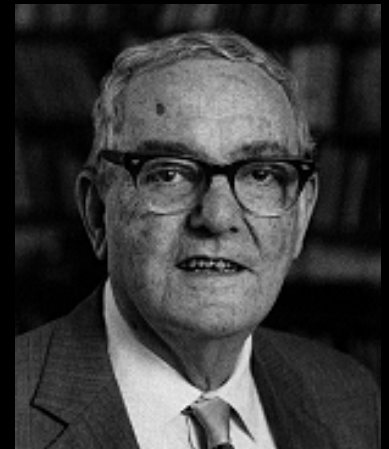
Alan Kay



Butler Lampson

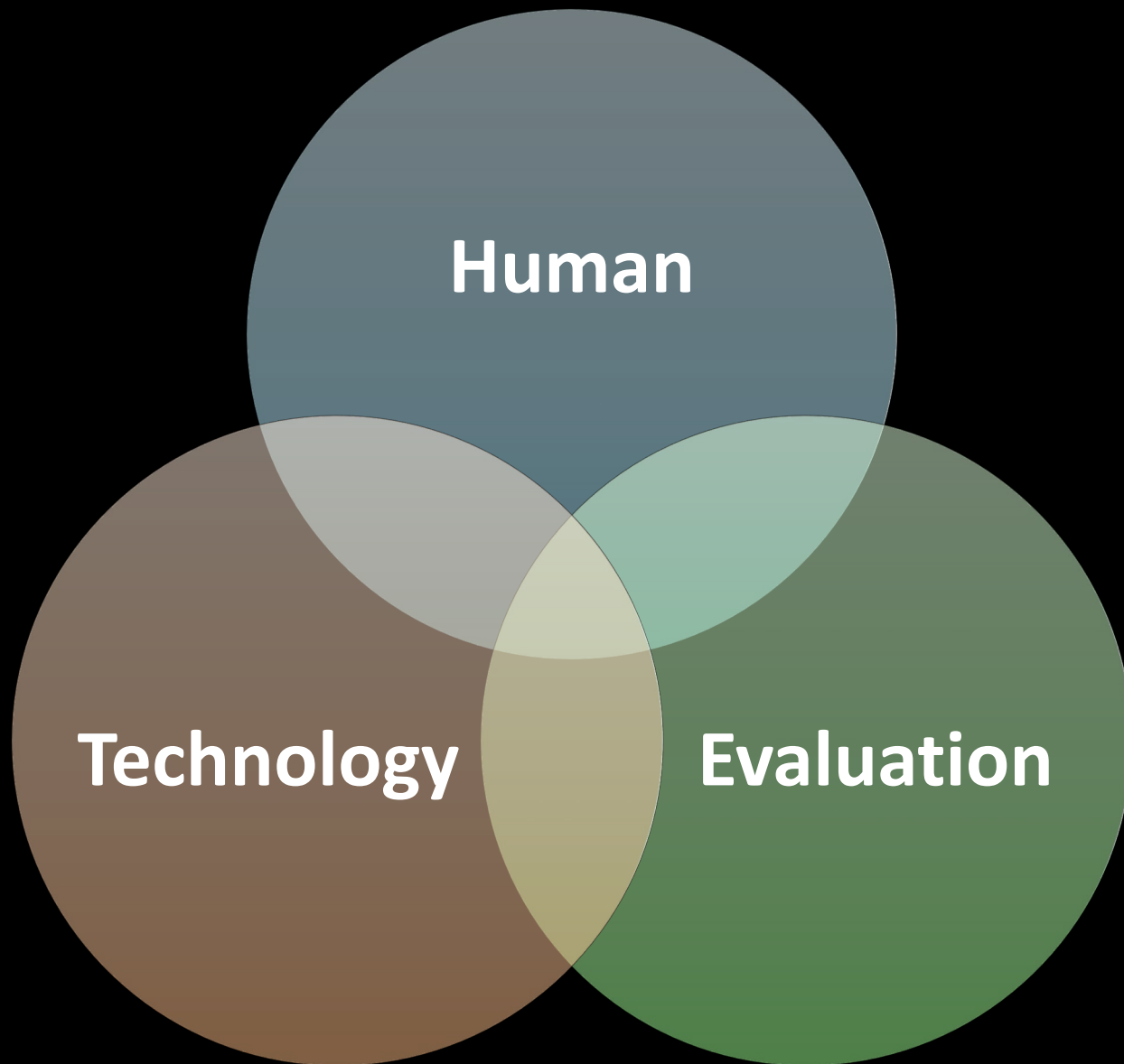


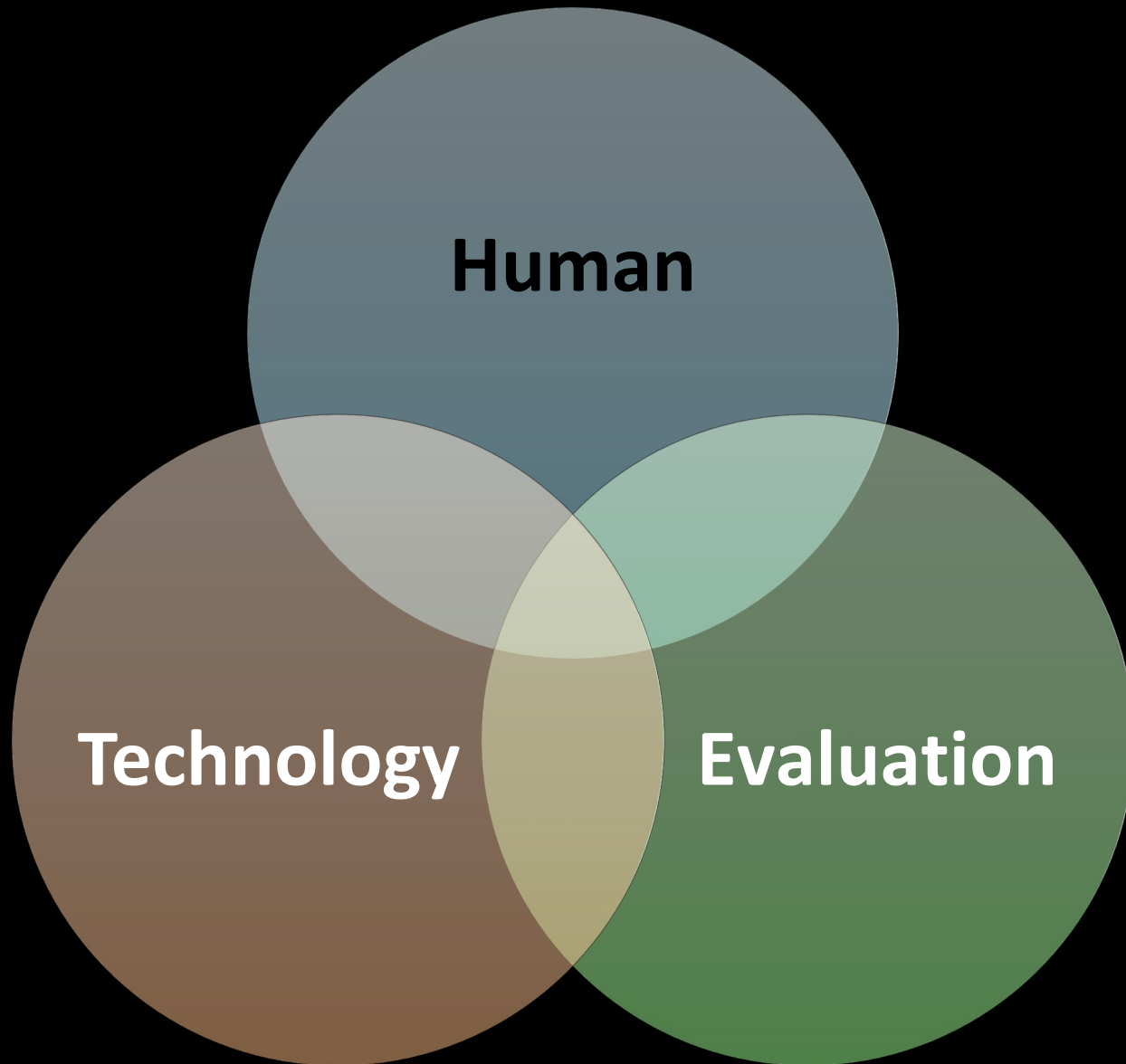
Allen Newell



Herbert Simon

Why Study HCl?

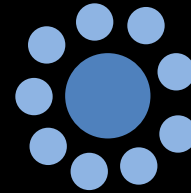
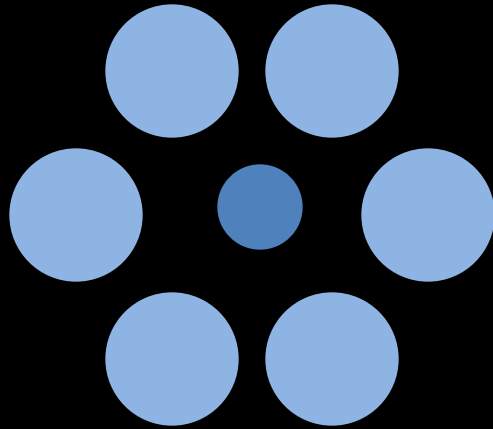




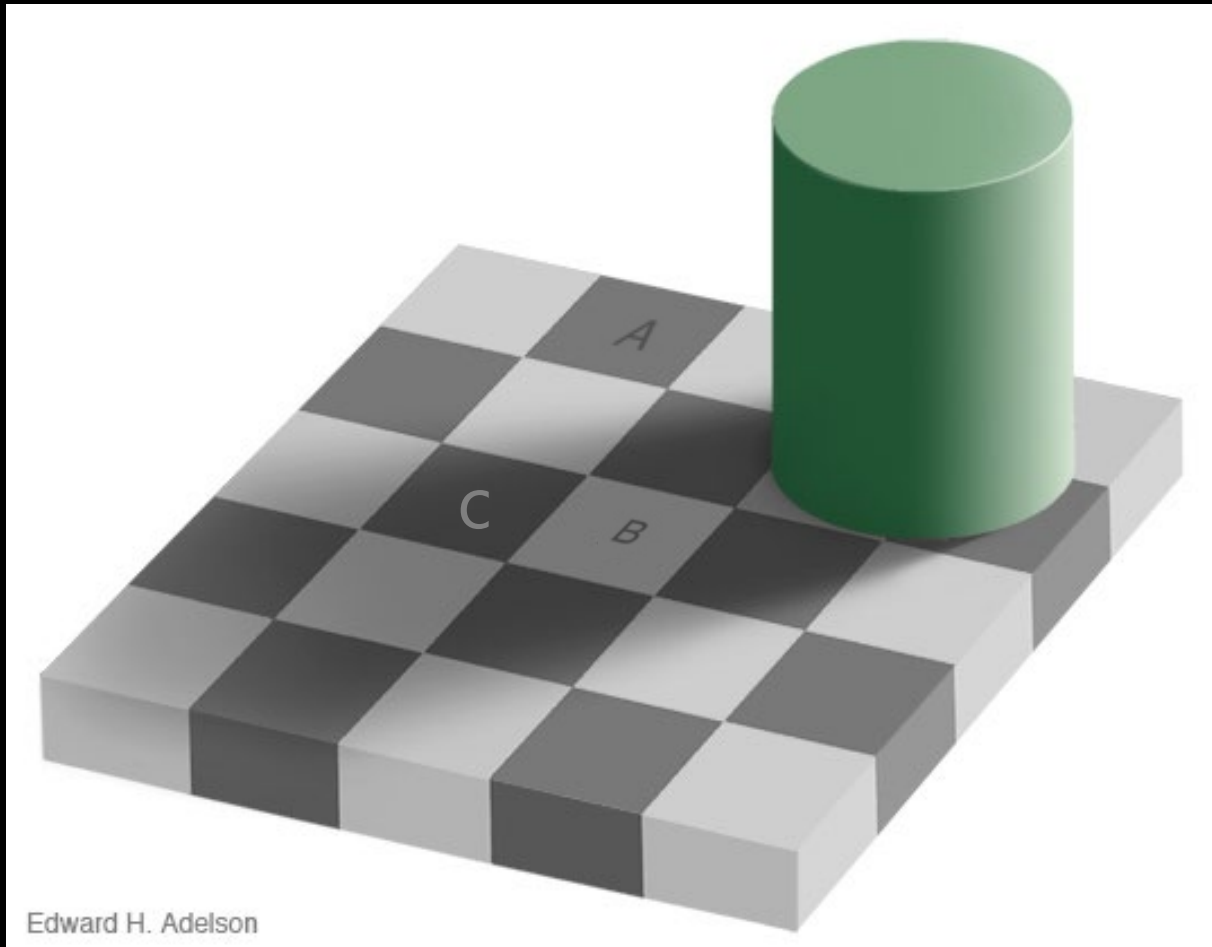
**Human does not behave
like a machine.**

Our Perception System is Biased.

Which Circle is Bigger?



Which Region is Darker?



Why Our Decision System is Biased

Bias toward certainty

**50% chance
of winning
\$200**

**100% chance
of winning \$90**

Bias toward uncertainty

**50% chance of
losing \$200**

**100% chance of
losing \$90**

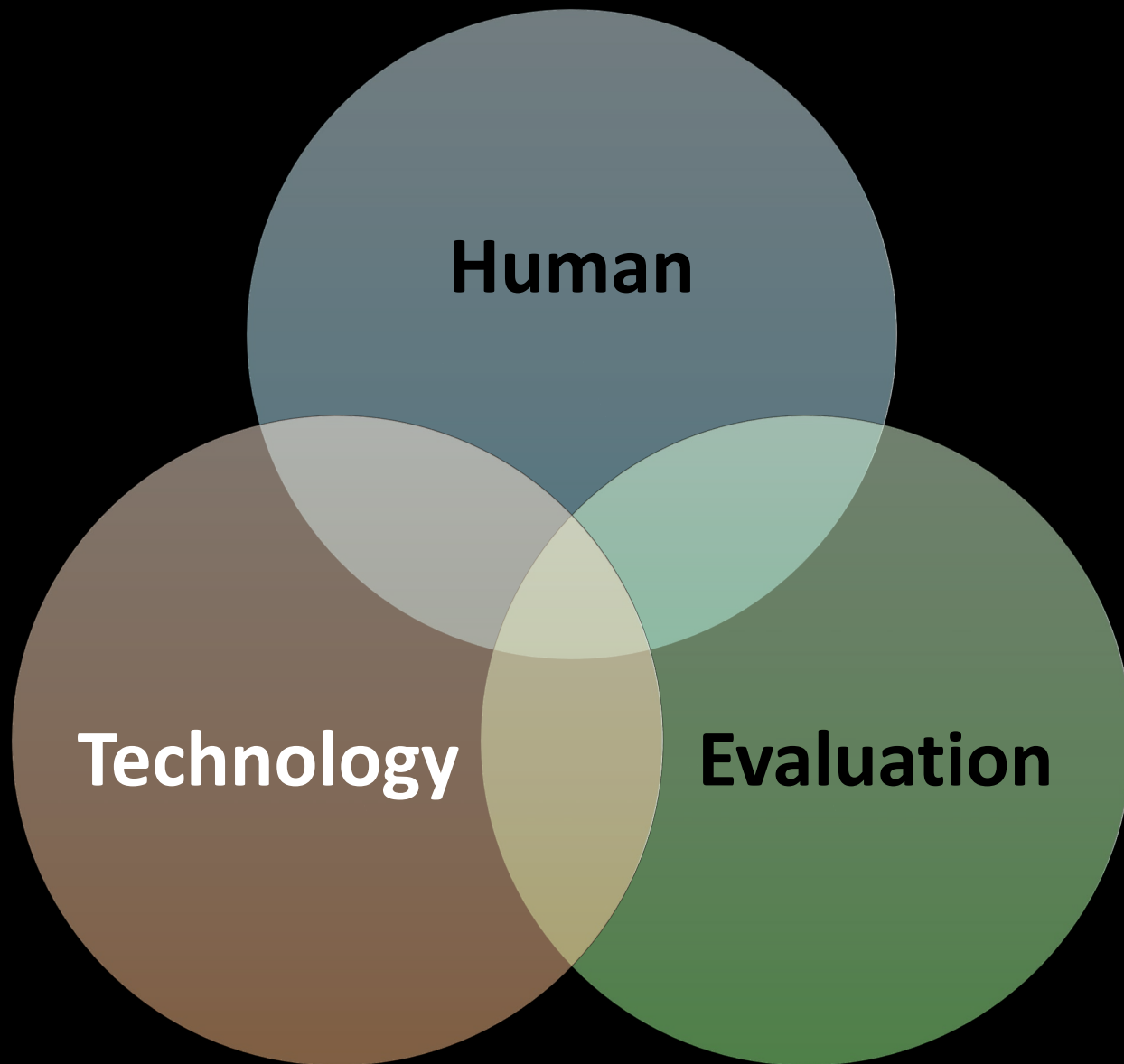
Peak-end Effect

- **Peak-End Rule**
 - The most intensive (peak) and ending moments play a dominant role

Which progress bar is faster?

Bar no. 2



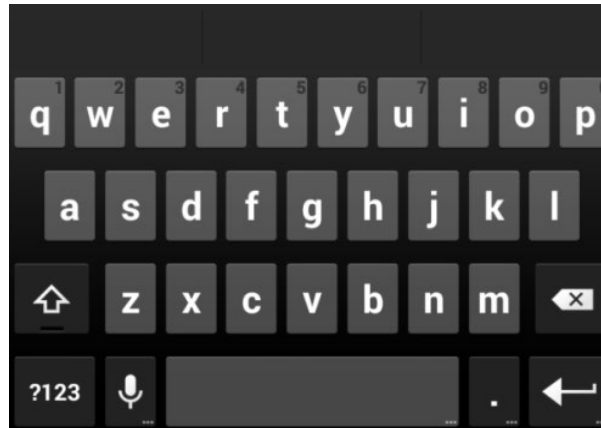


Smart Touch Keyboard

Typed Word

Keyboard Output

agsim



again

quivj



quick

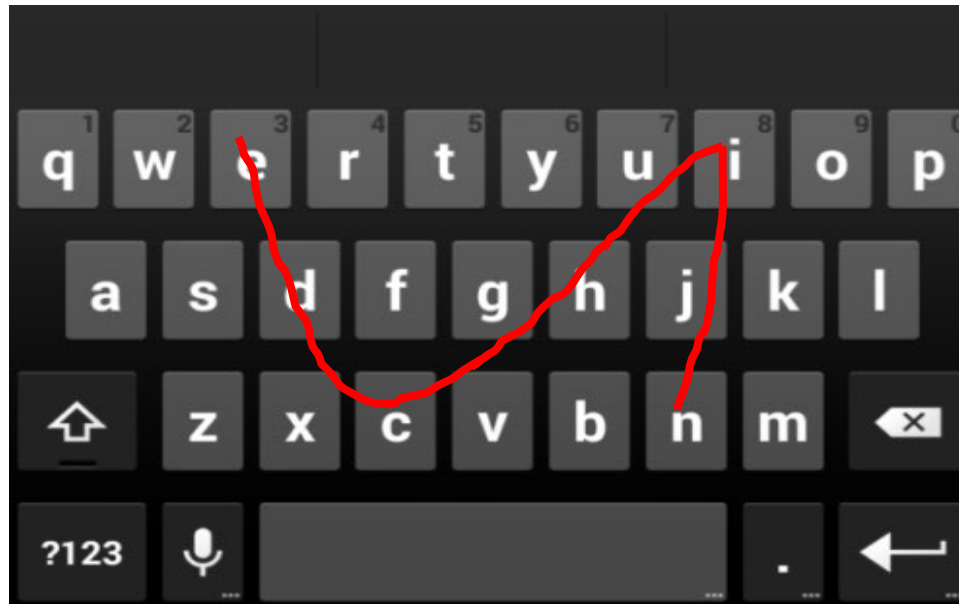
fav



favorite

Gesture Keyboard

Entering *nice*



Multi-Modal Interfaces

- *Multimodal* generally refers to an interface that can accept input from two or more combined modes

- Input Modalities

mouse

pen

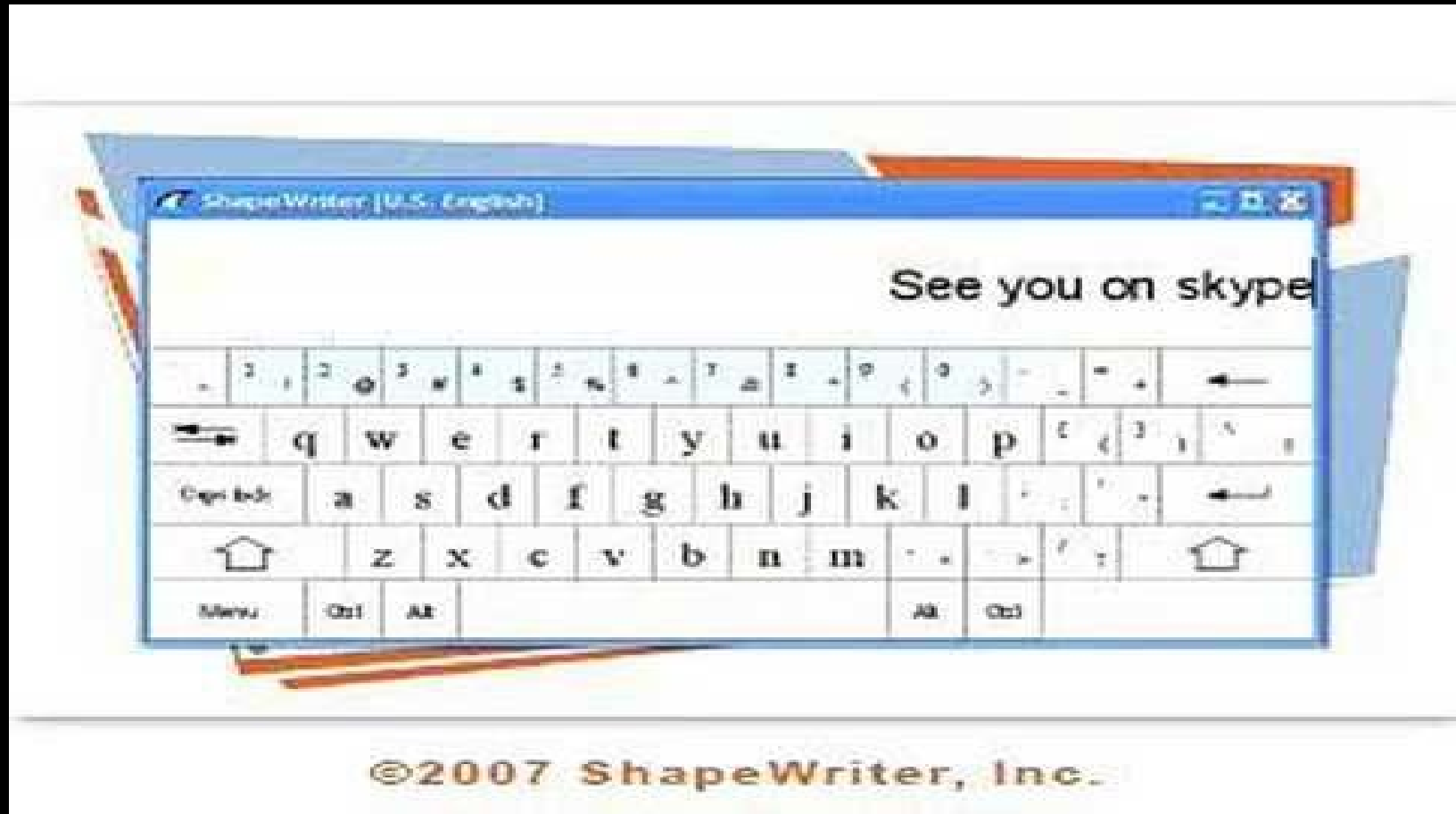
speech

non-speech audio

tangible object manipulation

gaze, posture, body-tracking

Example 1. ShapeWriting Technique

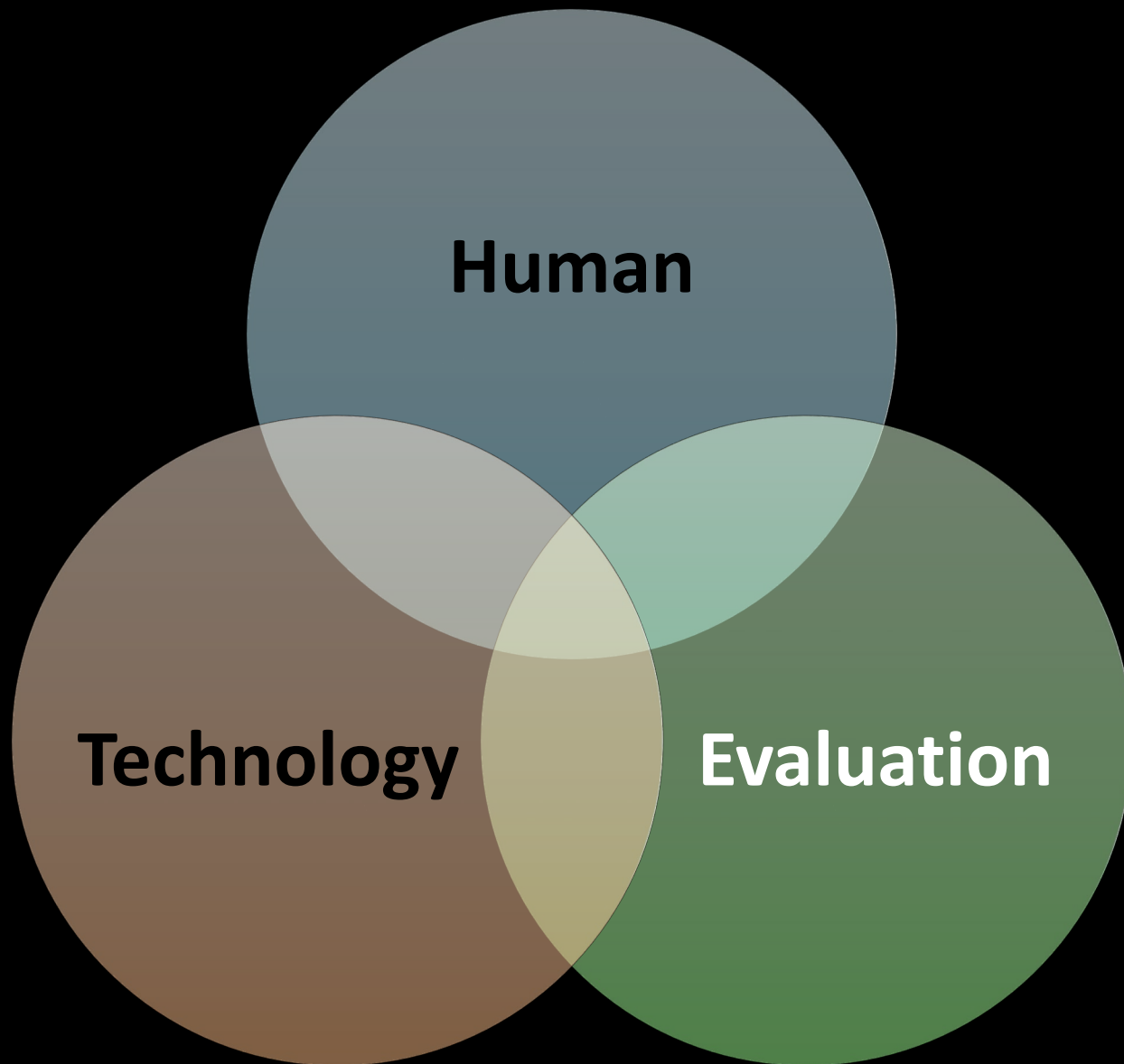


Per-Ola Kristensson and Shumin Zhai. 2004.
SHARK2: a large vocabulary shorthand writing system for pen-based computers. (UIST '04), 43-52.

Example 2. Mobile Sensing Techniques



Ken Hinckley, Jeff Pierce, Mike Sinclair, and Eric Horvitz. 2000.
Sensing techniques for mobile interaction (UIST '00). 91-100.



Heuristics

H2-1: Visibility of system status

H2-2: Match system and real world

H2-3: User control and freedom

H2-4: Consistency and standards

H2-5: Error prevention

H2-6: Recognition rather than recall

H2-7: Flexibility and efficiency of use

H2-8: Aesthetic and minimalist design

H2-9: Help users recognize, diagnose and recover from errors

H2-10: Help and documentation

Evaluation

- How to evaluate an interactive system or interface design?
- How to design an experiment?
- How to analyze the study data?
- How to present the study results?

Discount Usability Engineering

Cheap

- No special labs or equipment needed
- The more careful you are, the better it gets

Fast

- On order of 1 day to apply
- Standard usability testing may take a week

Easy to use

- Can be taught in 2-4 hours

Between vs. Within Subjects

Between subjects

- Each participant uses one condition
 - + Avoid carry-over learning effects
 - - Participants cannot compare conditions
 - - Need more participants

Within subjects

- All participants try all conditions
 - + Compare one person across conditions to isolate effects of individual diffs
 - + Requires fewer participants
 - - Fatigue effects
 - - Bias due to ordering/learning effects

A Glance of The Course

- Human Performance Modeling
- Input and Interaction Techniques
- Affordance, Conceptual Model and Design Principles
- Evaluation Techniques
- Interface Optimization
- AI-Powered Interaction

Administrivia

Course Info

https://www3.cs.stonybrook.edu/~xiaojun/courses/CSE518_2025.html

My Info

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Textbook

- No official textbook.
- Cutting edge research published in top HCI conferences (*ACM CHI* and *UIST*)
- Seminal books

Seminal Books

- *The Design of Everyday Things* (2013) by Don Norman
- *The Psychology of Human-Computer Interaction* (1983) by Stuart K. Card, Thomas P. Moran, and Allen Newell
- *Doing Psychology Experiments* (7th Edition, 2008) by David W. Martin

Communication Policy

- Most of your questions (homework, deadlines, projects, etc.) should be publicly asked on Piazza, so that other students can benefit from the answers.

<https://piazza.com/stonybrook/fall2025/cse518>

- Any personal email to the instructor or TA should include "[CSE518]" in the title.

Grading

- **20% Mid-term Exam.**

close-book exam.

- **20% Third-quarter Exam.**

close-book exam.

- **30% Three Individual Homework Assignments.**

- **25% Individual Project for Research Project**

Points breakdown for project: 25% =
execution (8%) + final report (8%) +
presentation (5%) + project description slides (1%) + project review (3%).

- **5% Participation**

In class discussions

Projects

Research Projects

- The “doing” part of the course
- A project related to your research is great
- Let me know if you do this

Questions?